A limit-periodic Schrödinger operator with mixed absolutely continuous and singular continuous spectrum

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Abstract

We describe a recursive construction of a bounded real potential on $\mathbb{Z}$, uniformly approximated by periodic potentials, for which the associated scalar discrete Schrödinger operator has nonzero absolutely continuous and singular continuous spectral subspaces and no eigenvalues. The mechanism combines adiabatic preservation of a local spectral density, exponential narrowing of a different spectral cluster, and exact local periodic repetitions. The quantitative point is that the protected Floquet energies have a margin of order the inverse new period, whereas the selected spectral bands have exponentially small widths.

Problem and attribution. This construction gives a negative answer to AIM problem 1021 at repository revision `b014a51815ff`, namely Question 10.1 in Damanik–Fillman [1]. The construction is submitted here by Matthew J. Colbrook. Periodic spectral formulas and the Gordon argument are standard and are identified below.

1 The construction theorem

For a bounded real sequence V, let

$$(H_V u)(n) = u(n-1) + u(n+1) + V(n)u(n)$$

on $\ell^2(\mathbb{Z})$. Write μ_V for the spectral measure of δ_0. A sequence is limit-periodic if it is a uniform limit of periodic sequences.

Theorem 1. *There is a bounded real limit-periodic potential V such that*

$$\mathcal{H}_{\mathrm{pp}}(H_V) = \{0\}, \qquad \mathcal{H}_{\mathrm{ac}}(H_V) \neq \{0\}, \qquad \mathcal{H}_{\mathrm{sc}}(H_V) \neq \{0\}.$$

More specifically, there are a compact set K of positive Lebesgue measure and a constant $c > 0$ such that $\mu_V \geq c\mathbf{1}_K(E)\,dE$. There is also a bounded interval I with endpoints in the resolvent set for which $\mathbf{1}_I(H_V) \neq 0$ and $\mathrm{Leb}(\sigma(H_V) \cap I) = 0$.

We construct periodic approximants W_j of periods p_j, with $p_j \mid p_{j+1}$. At each step a slow scalar offset is added to successive p_j-site cells. One set of energies remains elliptic throughout the slow cycle; its local spectral density is preserved on a slightly smaller compact set. A second, isolated spectral cluster encounters a long forbidden plateau at every energy; its new bands are exponentially narrow. A zero plateau at the center of the cycle preserves exact three-block repetitions.

2 Periodic formulas

For a real p-periodic W, put

$$T_v(E) = \begin{pmatrix} E - v & -1 \\ 1 & 0 \end{pmatrix}, \quad M_W(E) = T_{W(p-1)}(E) \cdots T_{W(0)}(E), \quad D_W(E) = \operatorname{tr} M_W(E).$$

The monodromy transfers $(u(0), u(-1))^{\mathsf{T}}$ to $(u(p), u(p-1))^{\mathsf{T}}$. Define the strictly elliptic set

$$\mathcal{E}_W = \{E : |D_W(E)| < 2\}.$$

The spectrum is $\{|D_W| \le 2\}$, a union of at most p bands. A closed Floquet gap may satisfy $|D_W| = 2$ inside the interior of the spectral set; thus strict ellipticity is the relevant notion for the construction.

On $\mathcal{E}_W$, the density of μ_W is

$$\rho_W(E) = \frac{|(M_W(E))_{12}|}{\pi\sqrt{4 - D_W(E)^2}}. \tag{2.1}$$

To check the index and normalization, write $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Bloch solutions normalized by $u_\pm(0) = 1$ satisfy $u_\pm(-1) = (\lambda_\pm - a)/b$, where $\lambda_\pm$ are the Floquet multipliers. Their Wronskian gives the diagonal Green function, and its boundary imaginary part gives (2.1). For $W = 0$, this is $1/(\pi\sqrt{4 - E^2})$; there is no extra period factor. See also [2, Chapter 7].

Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad R_\theta = e^{\theta J}.$$

On each open component of $\mathcal{E}_W$ choose smooth real $B(E) \in \mathrm{SL}(2, \mathbb{R})$ and a real phase lift $\theta(E)$ with

$$M_W(E) = B(E) R_{\theta(E)} B(E)^{-1}. \tag{2.2}$$

There is no topological issue since the component is an interval. For an explicit choice, set

$$\tau = -\operatorname{sgn}(b)\sqrt{1 - D_W^2/4}, \quad \beta = -b/\tau > 0, \quad t = (a - d)/(2\tau), \quad B = \begin{pmatrix} \sqrt{\beta} & 0 \\ t/\sqrt{\beta} & 1/\sqrt{\beta} \end{pmatrix},$$

and take $\cos\theta = D_W/2$, $\sin\theta = \tau$. The entry b cannot vanish at a real elliptic matrix. In particular,

$$(BJB^{-1})_{12} = -\beta, \qquad \rho_W = \beta/(2\pi). \tag{2.3}$$

The signed phase has a strictly negative energy derivative. Put $P_0 = \mathrm{I}$ and $P_r = T_{W(r-1)} \cdots T_{W(0)}$. Differentiating the product gives

$$M_W^{-1}(M_W)_E = -JC, \qquad C = \sum_{r=0}^{p-1} P_r^{\mathsf{T}} e_1 e_1^{\mathsf{T}} P_r.$$

In the coordinates of (2.2), the contribution from differentiating B is $R_\theta^{-1} L R_\theta - L$, where $L = B^{-1} B_E$. Multiplication by J and taking traces kills that term. Consequently

$$\theta_E = -\tfrac{1}{2} \operatorname{tr}(B^{\mathsf{T}} C B) < 0. \tag{2.4}$$

The matrix C is nonnegative and nonzero. Thus on compact subsets of an elliptic component, $-\theta_E$ is bounded above and below by positive constants.

3 The slow profile

Fix a smooth real 1-periodic function f with $|f| \le 1$, identically zero near $\mathbb{Z}$, and equal to $+1$ and -1 on two other intervals of positive length. Two disjoint smooth bumps of opposite signs provide such a function. All three plateau lengths are fixed.

For a p-periodic W, $\delta > 0$, and an integer N, define a $q = pN$-periodic potential by

$$\mathcal{S}_{N,\delta} W(mp + r) = W(r) + \delta f(m/N), \qquad 0 \le r < p, \quad m \in \mathbb{Z}. \tag{3.1}$$

Its distance from W is at most δ. The transfer matrix across its mth old cell is exactly $M_W(E - \delta f(m/N))$.

4 Adiabatic preservation of a bulk density

Slow-deformation estimates of this kind appear in Avila [3, Lemmas 2.11–2.12]. We give the version needed here, including a quantitative margin from every new discriminant edge.

Lemma 2. *Let K be a finite union of compact intervals with $K + [-\delta, \delta] \Subset \mathcal{E}_W$. For every sufficiently small fixed $\eta > 0$ and all sufficiently large N, there is a finite union of compact intervals $G_N \subset K$ such that, for $W_N = \mathcal{S}_{N,\delta} W$ and $q = pN$,*

$$\mathrm{Leb}(K \setminus G_N) \le C\eta, \tag{4.1}$$

$$\sup_{G_N} |\rho_{W_N} - \rho_W| \le C/(N\eta^2), \tag{4.2}$$

$$\mathrm{dist}(G_N, \mathbb{R} \setminus \mathcal{E}_{W_N}) \ge c\eta^2/q. \tag{4.3}$$

The positive constants may depend on W, p, δ, K, f, but not on N or η. The threshold for N may depend on η.

Proof. Choose a slightly enlarged neighborhood U of K with $\overline{U} + [-\delta, \delta] \Subset \mathcal{E}_W$. Every shifted component stays in one old elliptic component. Write

$$A(s, E) = M_W(E - \delta f(s)) = B(s, E) R_{\theta(s,E)} B(s, E)^{-1},$$

where $B(s, E) = B_W(E - \delta f(s))$ is chosen as in (2.2). Then $B(1, E) = B(0, E) = B_W(E)$. All required derivatives are bounded on the compact parameter set.

Set $h = N^{-1}$ and decompose $B^{-1} B_s = gJ + S$, where S is symmetric and traceless. On that two-dimensional real matrix space, Ad_{R_θ} is a rotation through 2θ. Since $|\sin\theta|$ is bounded away from zero, the equation

$$(\mathrm{Ad}_{R_\theta} - \mathrm{I}) X = S$$

has a unique smooth bounded solution in that space. With $P(s, E) = e^{hX(s,E)}$ and coordinates $U_m = B(mh, E) P(mh, E) Z_m$, one old-cell step becomes

$$\begin{aligned} Z_{m+1} &= P(s + h)^{-1} B(s + h)^{-1} B(s) R_\theta P(s) Z_m \\ &= (R_{\theta - hg} + O(h^2)) Z_m. \end{aligned} \tag{4.4}$$

Indeed $B(s + h)^{-1} B(s) = \mathrm{I} - h(gJ + S) + O(h^2)$, and the first-order nonrotational term is $\mathrm{Ad}_{R_\theta} X - X - S = 0$. The remainder is uniform.

Products of at most N factors in (4.4) are uniformly bounded by $(1 + Ch^2)^N$. Telescoping against their rotation parts gives error $O(Nh^2) = O(N^{-1})$. Define

$$\Phi_N(E) = \sum_{m=0}^{N-1} (\theta(m/N, E) - N^{-1} g(m/N, E)).$$

The full monodromy is therefore

$$M_{W_N}(E) = B_W(E)(R_{\Phi_N(E)} + O(N^{-1}))B_W(E)^{-1}. \tag{4.5}$$

In fact $X(0,E) = X(1,E) = 0$ because the profile is constant near those endpoints, so the endpoint correction is trivial. All partial transfers inside the q-site cell and their inverses are bounded independently of N. The at most p extra one-site factors within an old cell do not change this.

By (2.4), for all sufficiently large N,

$$c_1 N \le -\Phi_N'(E) \le C_1 N$$

on each component of U. Set $G_N = \{E \in K : |\sin \Phi_N(E)| \ge \eta\}$. The monotone phase change of variables proves (4.1): in a phase interval of length L, the set $|\sin\phi| < \eta$ has length at most $C\eta(L+1)$. There are only finitely many crossings, so G_N is a finite union of compact intervals.

With $\beta = -(B_W J B_W^{-1})_{12} > 0$, (4.5) gives

$$D_{W_N} = 2\cos\Phi_N + O(N^{-1}), \qquad (M_{W_N})_{12} = -\beta \sin\Phi_N + O(N^{-1}).$$

On G_N, when $N^{-1} \ll \eta^2$, the trace lies strictly between -2 and 2. With $s = |\sin\Phi_N| \ge \eta$,

$$\sqrt{4 - D_{W_N}^2} = 2s + O((Ns)^{-1}).$$

Substitution in (2.1) gives $\rho_{W_N} = \beta/(2\pi) + O((N\eta^2)^{-1})$, proving (4.2).

Differentiate the q-site transfer product. It is a sum of q terms, each bounded uniformly by the bounds on left and right partial transfers. Thus $|D'_{W_N}| \le Cq$ on U. On G_N, the trace has margin $2 - |D_{W_N}| \ge c\eta^2$. For each $E \in G_N$, the interval of radius $c'\eta^2/q$ about E lies in its component of U for all large N. Choosing $c' > 0$ small enough, the derivative bound shows that this whole interval remains strictly elliptic. This proves (4.3), including when U is disconnected. □

Remark 3. The margin is measured from every point with $|D_{W_N}| \ge 2$, including closed gaps. Only the retained cores are used. No convergence assertion across the moving band edges, and no bound on spectral mass in moving exceptional sets, is required.

5 Exponentially narrow target bands

Lemma 4. *Let I be a compact interval and suppose for some $d > 0$ that*

$$\max\{\operatorname{dist}(E, \sigma(H_{W+\delta})), \operatorname{dist}(E, \sigma(H_{W-\delta}))\} \ge d \quad (E \in I). \tag{5.1}$$

Every portion in I of a Floquet band of $W_N = \mathcal{S}_{N,\delta}W$ has length at most Ce^{-cN}. Consequently

$$\operatorname{Leb}(\sigma(H_{W_N}) \cap I) \le C(pN)e^{-cN}.$$

Constants are independent of N. If the endpoints of I are in spectral gaps, the portions are complete bands.

Proof. Each ± 1 plateau contains a consecutive run of old cells of total length $L \ge aq$, where $q = pN$ and $a > 0$ is fixed, for large N. On it the potential is exactly $W + \delta$ or $W - \delta$.

For a Floquet energy $E = E_k(\vartheta) \in I$, $0 < \vartheta < \pi$, normalize the Bloch solution by

$$u(n+q) = e^{i\vartheta}u(n), \qquad \sum_{n=0}^{q-1} |u(n)|^2 = 1.$$

Then $|u(n)| \le 1$ for all n. Choose the plateau operator H_U whose spectrum is at least d from E. Cutting u off to that plateau creates at most four boundary sources for $H_U - E$, each of modulus at most one.

The elementary Combes–Thomas bound reads

$$|\langle \delta_m, (H_U - E)^{-1} \delta_n \rangle| \le \frac{2}{d} e^{-\gamma |m-n|}, \qquad \gamma = \operatorname{arsinh}(d/4).$$

To see it, conjugate by a bounded clipped exponential weight: the two hopping terms change in norm by at most $2 \sinh \gamma = d/2$, and the conjugated resolvent has norm at most $2/d$. Thus at two adjacent middle sites of the plateau,

$$|u(r)|, \ |u(r+1)| \le \frac{8}{d} e^{-\gamma(L/2-2)}.$$

The current $2 \operatorname{Im}(u(n+1)\overline{u(n)})$ is independent of n. Hellmann–Feynman differentiation of the boundary twist identifies it, up to sign, with $E_k'(\vartheta)$ under the stated normalization. Hence

$$|E_k'(\vartheta)| \le 128 d^{-2} e^{4\gamma} e^{-\gamma L}.$$

Scalar Floquet branches are simple and monotone on $(0, \pi)$. Simplicity follows from the distinct multipliers $e^{\pm i\vartheta}$. The current cannot vanish there, since that would make the Bloch solution and its conjugate linearly dependent. Integrating over $[0, \pi]$ proves the band-portion estimate. There are at most q bands. □

Lemma 5. *If S is a union of at most p compact intervals and $r > p \operatorname{Leb}(S)$, there is $\delta \in (r, 2r)$ such that $(S + \delta) \cap (S - \delta) = \varnothing$.*

Proof. If the component lengths are $\ell_1, \dots, \ell_m$, then

$$\operatorname{Leb}(S - S) \le \sum_{i,k} (\ell_i + \ell_k) = 2m \operatorname{Leb}(S) \le 2p \operatorname{Leb}(S) < 2r.$$

Therefore $(2r, 4r)$ contains a point outside the compact set $S - S$. Take that point to be 2δ. □

Suppose I isolates $S = \sigma(H_W) \cap I$ from the rest of the spectrum and δ is smaller than the isolation margin. Disjointness of $S + \delta$ and $S - \delta$ then implies (5.1) for some $d > 0$: the two full shifted spectra have no common point in I, so the continuous maximum of the two distances has a positive minimum on compact I. The value of d may depend on the chosen shift.

6 Concrete initial data

Take the period-three background

$$W_0 = (100, 0, 0), \qquad p_0 = 3, \qquad D_0(E) = E^3 - 100E^2 - 3E + 100.$$

Fix

$$I = [99, 101], \qquad K_0 = [0.989, 0.991], \qquad \delta_1 = 10^{-3}.$$

The factorizations

$$D_0(E) + 2 = (E - 1)(E^2 - 99E - 102), \qquad D_0(E) - 2 = (E + 1)(E^2 - 101E + 98)$$

show that the upper band S_0 lies in $(100, 100.1)$, while the other two bands lie in $[-2, 2]$. Since $D_0' \geq 9997$ on the upper interval,

$$\mathrm{Leb}(S_0) \leq 4/9997 < 0.000401 < 2\delta_1.$$

The two translates $S_0 \pm \delta_1$ are disjoint, giving the barrier hypothesis on I.

For $E \in K_0$ and $|t| \leq \delta_1$, one has $E - t \in [0.988, 0.992]$. On this interval $D_0' < 0$, and

$$D_0(0.988) = 0.386030272, \qquad D_0(0.992) = -0.406208512.$$

Hence $K_0 + [-\delta_1, \delta_1] \Subset \mathcal{E}_{W_0}$. Let $m_0 = \min_{K_0} \rho_{W_0} > 0$.

Every later amplitude will be at most one quarter of the preceding one. Therefore the total norm change from W_0 is at most $(4/3)10^{-3} < 0.002$. The endpoints of I stay in fixed spectral gaps, the upper cluster remains nonempty, and the other spectrum stays far below I.

7 Induction with explicit parameter order

Construct $W_j, p_j, K_j, \delta_{j+1}$ and write

$$S_j = \sigma(H_{W_j}) \cap I, \qquad g_j = \mathrm{dist}(K_j, \mathbb{R} \setminus \mathcal{E}_{W_j}).$$

Fix positive tolerances α_j, ϵ_j with

$$\sum_{j \geq 1} \alpha_j < \tfrac{1}{2} \mathrm{Leb}(K_0), \qquad \sum_{j \geq 1} \epsilon_j < \tfrac{1}{2} m_0.$$

At the start of stage j, the data $W_{j-1}, p_{j-1}, K_{j-1}, \delta_j$ are fixed. The shifted bulk is strictly elliptic and the shifted target clusters are disjoint. This holds at $j = 1$ by the explicit initial data.

For every candidate $W_N = \mathcal{S}_{N,\delta_j} W_{j-1}$, the already chosen amplitudes give

$$\|W_N - W_0\|_\infty \leq \sum_{k=1}^{j} \delta_k \leq \tfrac{4}{3}\delta_1 < 0.002.$$

The same bound holds for $W_{j-1} \pm \delta_j$. Their spectra are contained in $[-2.002, 2.002] \cup [99.998, 100.102]$. This makes the fixed isolation of I a finite-stage invariant.

Apply Lemma 2. Choose $\eta_j > 0$ first so that the removed measure is less than α_j. Let N tend to infinity, keeping those data and η_j fixed. The candidate outputs

$$W_N = \mathcal{S}_{N,\delta_j} W_{j-1}, \quad p_N = p_{j-1} N, \quad K_N = G_N$$

satisfy, for all large N,

$$\begin{aligned} \mathrm{Leb}(K_{j-1} \setminus K_N) &< \alpha_j, \\ \sup_{K_N} |\rho_{W_N} - \rho_{W_{j-1}}| &< \epsilon_j, \\ g_N := \mathrm{dist}(K_N, \mathbb{R} \setminus \mathcal{E}_{W_N}) &\geq \kappa_j / p_N, \qquad (7.1) \\ \mathrm{Leb}(S_N) &\leq C_j p_N e^{-c_j N}. \qquad (7.2) \end{aligned}$$

The constants $\kappa_j, C_j, c_j > 0$ are independent of the new N.

For each candidate define

$$r_N = \min\{g_N/8,\ 2^{-j-1}/p_N,\ \delta_j/8\}. \qquad (7.3)$$

Since $p_N \geq 1$, (7.1) gives

$$r_N \geq b_j/p_N, \qquad b_j = \min\{\kappa_j/8, 2^{-j-1}, \delta_j/8\} > 0.$$

Meanwhile $p_N \operatorname{Leb}(S_N) \leq C_j p_N^2 e^{-c_j N}$. Because $p_N = p_{j-1}N$, choose $N = N_j$ large enough that

$$p_N \operatorname{Leb}(S_N) < r_N. \tag{7.4}$$

This is exactly the comparison $N^3 e^{-c_j N} \to 0$. Also require each N_j to exceed a fixed N_*, chosen below for the repetitions.

Set $W_j = W_{N_j}$, $p_j = p_{N_j}$, $K_j = K_{N_j}$, and $r_j = r_{N_j}$. Lemma 5 gives $\delta_{j+1} \in (r_j, 2r_j)$ separating the target translates. Its upper bounds are

$$\delta_{j+1} < g_j/4, \qquad \delta_{j+1} < 2^{-j}/p_j, \qquad \delta_{j+1} < \delta_j/4. \tag{7.5}$$

The first ensures $K_j + [-\delta_{j+1}, \delta_{j+1}] \Subset \mathcal{E}_{W_j}$. The fixed target isolation ensures the barrier hypothesis. All constants for the following stage are now fixed before N_{j+1} is chosen. There is no circular choice of parameters.

Since $\|W_j - W_{j-1}\|_\infty \leq \delta_j$, these approximants converge uniformly to a real bounded V. Their periods are nested and tend to infinity. Thus V is limit-periodic and $\|V\|_\infty < 100.002$. Equivalently, the resulting potential is

$$V(n) = W_0(n) + \sum_{j \geq 1} \delta_j f\left(\frac{\lfloor n/p_{j-1} \rfloor}{N_j}\right).$$

8 Surviving absolutely continuous spectrum

The compact set $K_\infty = \bigcap_{j \geq 0} K_j$ has positive Lebesgue measure, by the summable losses. For $E \in K_\infty$,

$$\rho_{W_j}(E) \geq m_0 - \sum_{k=1}^{j} \epsilon_k > m_0/2.$$

Thus every approximating spectral measure dominates the same nonzero absolutely continuous measure

$$\nu = (m_0/2)\mathbf{1}_{K_\infty}(E)\, dE.$$

Uniform convergence of potentials gives norm convergence of their operators and weak convergence $\mu_{W_j} \Rightarrow \mu_V$. Pass to the limit against each nonnegative continuous test function in $\mu_{W_j} \geq \nu$ to obtain $\mu_V \geq \nu$. Consequently $\mathcal{H}_{\mathrm{ac}}(H_V) \neq \{0\}$.

Remark 6. Nonzero absolutely continuous spectrum has an independent verification through reflectionless compactness. Each periodic W_j is reflectionless on the same set K_∞. The bounded reflectionless class on this fixed set is closed under coefficientwise convergence; hence V is reflectionless there. Remling [4, Proposition 4.1(d)] gives this compactness on the fixed set K_∞, with the common coefficient bound. Part (b) gives absolutely continuous spectrum for the half-line restrictions: their Weyl functions have positive boundary imaginary parts almost everywhere on K_∞. The formula for the full-line diagonal Green function in terms of these Weyl functions then gives a nonzero absolutely continuous component for H_V; see also [2, Chapter 2]. This optional alternative is not used in the direct measure proof. The direct measure argument above additionally supplies the stated lower bound on the chosen local spectral measure.

9 Surviving singular spectrum

Put $t_j = \|V - W_j\|_\infty$. Geometric decay and (7.5) give

$$t_j \le \tfrac{4}{3}\delta_{j+1}, \qquad p_j t_j \le \tfrac{4}{3}\, 2^{-j} \longrightarrow 0.$$

Hausdorff spectral continuity, together with the fixed gaps at the endpoints of I, gives for large j

$$\sigma(H_V) \cap I \subset (S_j)_{t_j},$$

where $(S_j)_{t_j}$ is the closed t_j-neighborhood. There are at most p_j intervals in S_j, so

$$\mathrm{Leb}(\sigma(H_V) \cap I) \le \mathrm{Leb}(S_j) + 2p_j t_j \longrightarrow 0.$$

Here $\mathrm{Leb}(S_j) \to 0$ also follows directly from (7.4)–(7.3), which give $\mathrm{Leb}(S_j) < 2^{-j-1}/p_j^2$.

The cluster remains nonempty: the initial spectrum has a point in $(100, 100.1)$ and the total norm change is less than 0.002. Equivalently its isolated nonzero spectral projection persists under this perturbation. Therefore $\mathbf{1}_I(H_V) \neq 0$, and its range is entirely singular because it is supported on a null set.

In fact the same vector δ_0 sees both spectral types. Every modulation vanishes at the origin, so $V(0) = 100$ and

$$\int (E - 100)^2 \, d\mu_V(E) = \|(H_V - 100)\delta_0\|^2 = 2.$$

The spectrum outside I lies in $[-2.002, 2.002]$. Consequently

$$\mu_V(I) \ge 1 - \frac{2}{97.998^2} > 0.99979.$$

Thus μ_V has positive mass on the null spectral set inside I, in addition to its absolutely continuous component on K_∞.

10 Exact repetitions exclude eigenvalues

Choose $a_0 > 0$ so that $f(s) = 0$ whenever $\mathrm{dist}(s, \mathbb{Z}) \le a_0$. Take $N_* \ge 3$ with $1/N_* < a_0$.

Fix j. On $[-p_j, 2p_j - 1]$ the old-cell indices in the next update are $-1, 0, 1$, so that update vanishes there. Every later update also vanishes there: its cell length is at least $3p_j$, and the indices are only $-1, 0$. Hence

$$V(n) = W_j(n) \quad (-p_j \le n \le 2p_j - 1). \tag{10.1}$$

For a solution $H_V u = Eu$, write $U(n) = (u(n), u(n-1))^{\mathsf{T}}$ and let A_j transfer across the middle block. Then

$$U(p_j) = A_j U(0), \quad U(2p_j) = A_j^2 U(0), \quad U(-p_j) = A_j^{-1} U(0).$$

Cayley–Hamilton gives

$$A_j^2 - (\mathrm{tr}\, A_j) A_j + \mathrm{I} = 0, \qquad A_j + A_j^{-1} = (\mathrm{tr}\, A_j)\mathrm{I}.$$

If $|\,\mathrm{tr}\, A_j| \le 1$, use the first identity; otherwise use the second. In either case,

$$\max\{\|U(-p_j)\|, \|U(p_j)\|, \|U(2p_j)\|\} \ge \tfrac{1}{2}\|U(0)\|.$$

Since $p_j \to \infty$, no nonzero solution is square summable. This is the exact three-block Gordon argument; no rapid *uniform* approximation rate is assumed. See also [1, Section 6.2].

Thus $\mathcal{H}_{\mathrm{pp}}(H_V) = \{0\}$. The nonzero singular projection found above is therefore singular continuous. Together with the absolutely continuous component, this proves Theorem 1.

Scope

The construction supplies a nonzero ac submeasure on a compact set of positive measure, and a nonzero purely sc restriction to the isolated interval I. It does not classify every energy outside I or assert that the protected set contains an interval. Neither assertion is needed for failure of purity.

References

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